VETLINK VETERINARY SYSTEM TESTING & VERIFICATION GUIDE

Clinical AI Dermatology Assistant: Full Operational Test Protocol & Quality Audit

Document Version: 2.4.0 (Production Candidate)	**Target Audience:** Thesis Panelists, Clinicians, Field Testers
Repository Branch: feature/clinical-workflow-and-data-integrity	**Date of Evaluation:** November 2026
Compliance: RA 8485 / Zero-Emoji Protocol / Offline-First PWA	**Evaluation Scope:** 12 System Modules & Clinical Workflows

1. Executive Overview & Purpose of This Guide

This verification document provides a standardized clinical and technical testing protocol for the Vetlink Veterinary AI Dermatology Progressive Web Application. Designed for thesis defense demonstrations, veterinary clinical evaluators, and academic panel audits, this guide outlines exact step-by-step procedures, preconditions, expected outcomes, and audit checklists across all core functional domains. Every module adheres strictly to clinical safety regulations, ensuring diagnostic decision support never supersedes attending veterinary judgment.

2. System Architecture & Test Environment Summary

Tier	Component	Specification	Key Role
Client PWA	React 18 + Vite	IndexedDB + ServiceWorker	Offline-first execution, fast patient intake, local prescription caching
Edge AI	ONNX Runtime Web	MobileNetV3 quantized	Sub-second on-device skin lesion inference without active internet
Backend API	FastAPI (Python 3.10+)	PyTorch / Pillow / SMTPLib	200-class domestic animal verification, severe scab safeguard, email dispatch
Cloud DB	Aiven MySQL Cloud	Encrypted SSL Transport	Multi-device sync, institutional audit trail, case synchronization

3. Clinical UI Refinement Breakdown (Audit of 4 Target Areas)

During clinical workflow evaluation, four interface deficiencies were identified and resolved in the Patients and Treatment modules. The following matrix summarizes the before-and-after implementation verified in the current production release:

Target Component	Previous State (Deficiency)	Resolved State (Production Implementation)
Diagnosis Selection (View Case & Override)	Unorganized flat dropdown with all species diagnoses intermixed into one long list, creating clinician fatigue.	Grouped **optgroup** dropdown segmented into Canine Protocols (Aso), Feline Protocols (Pusa), Avian Protocols (Ibon), and Rabbit Protocols (Kuneho), plus custom condition override.
Healing Stage Evaluation (Follow-up Recheck)	Heavy, unstyled black circular radio dots embedded within plain rectangular white cards.	Four distinct clinical severity cards with subtle colored borders and micro-indicators: Green (Resolved), Amber (Responding), Orange (Refractory), and Red (New Complication).

Clinical Triage & Action (Diagnostic Review)	Asymmetric 3-and-1 layout leaving empty grid gaps, cluttered with oversized dark radio blobs.	Symmetric, balanced **2x2 responsive grid** featuring soft status backgrounds, color-coded badges, and clean indicator micro-dots.
Treatment Protocol (Clinical Guideline)	Static text blocks that prevented veterinarians from adjusting medication dosages or adding personalized steps.	**100% Attending Veterinarian Editable** protocol: editable text inputs, dynamic step addition (+ Add Guideline Step), step deletion, and guideline reset.

4. Step-by-Step System Testing Protocols (12 Clinical Test Cases)

The following test cases must be executed systematically during clinical validation and defense demonstrations.

TC-01: Mobile File Upload vs Camera Invocation (Media Picker Test)	Audit Result: [PASS]
Clinical Objective: Verify that tapping 'Upload File' on mobile devices presents the native gallery/file picker rather than forcefully launching the camera.	**Preconditions:** Open Vetlink PWA on an Android or iOS mobile browser (or responsive emulator).
Execution Steps: 1. Navigate to 'New Patient Intake' or 'Treatment Scan'. 2. Tap the 'Upload File' button. 3. Observe device prompt behavior.	**Expected Clinical Outcome:** The device opens its native media library/file manager allowing selection of pre-existing photos. The camera is not forcefully launched unless the dedicated 'Open Camera' button is selected.

TC-02: Image Upload Percentage Indicator Test	Audit Result: [PASS]
Clinical Objective: Confirm real-time visual progress feedback during client-side image processing and payload preparation.	**Preconditions:** Select a high-resolution dermatological lesion image (>3 MB).
Execution Steps: 1. In New Patient or Treatment page, upload the high-resolution image. 2. Observe the upload status container.	**Expected Clinical Outcome:** A dynamic progress bar displays percentage progression (0% to 100%) during pre-processing, compression, and verification, preventing user confusion.

TC-03: Offline Demo Mode & Instant Session Termination	Audit Result: [PASS]
Clinical Objective: Verify that switching users or exiting demo mode terminates immediately without blocking on network timeouts.	**Preconditions:** Log in using the Demo Account credentials in an offline or airplane mode environment.
Execution Steps: 1. Navigate to the sidebar or Settings page. 2. Click 'Exit Demo Mode' or 'Log Out'. 3. Verify redirect to the login screen.	**Expected Clinical Outcome:** Session storage and authentication tokens are immediately cleared. The user is redirected to the login screen in less than 200ms with zero network hanging.

TC-04: Domestic Animal Skin Verifier & Severe Scab Safeguard	Audit Result: [PASS]
Clinical Objective: Ensure non-animal objects (blue screen, vehicle, food) are rejected while true severe canine/feline lesions are accepted.	**Preconditions:** Backend FastAPI running with PyTorch MobileNetV3 verifier.
Execution Steps: 1. Submit a synthetic blue screen or non-animal picture (e.g. keyboard, car). 2. Observe verifier rejection. 3. Submit a heavily crusted canine mange lesion.	**Expected Clinical Outcome:** Non-animal inputs are rejected with HTTP 422 ('Image Rejected: Not Identified as an Animal'). Heavily crusted lesions trigger the Severe Scab Safeguard (disease probability >= 60%) and proceed safely.

TC-05: Out-of-Dataset Diagnostic Safety Notice (Confirmatory Diagnostics)	Audit Result: [PASS]
Clinical Objective: Verify clinical disclosure that non-dataset conditions (e.g. Feline Acne, Malassezia) require confirmatory laboratory testing.	**Preconditions:** Access the New Patient Intake and AI Diagnostic pages.
Execution Steps: 1. Observe the diagnostic guidance banner below the AI inference card. 2. Check the recommendation text for laboratory protocols.	**Expected Clinical Outcome:** Prominently displays: 'Clinical Scope Notice: Diagnoses outside core dataset require confirmatory laboratory workup (Skin Scraping, Cytology, or DTM Culture)'.

TC-06: Clinical Triage 2x2 Grid & Action Status Selection	Audit Result: [PASS]
Clinical Objective: Verify balanced 2x2 grid rendering and status persistence in Case Review.	**Preconditions:** Open 'Patients' directory and click 'View Case' on any active consultation.

Execution Steps: 1. Scroll to 'Clinical Triage & Action Status'. 2. Toggle between 'Review Results', 'Needs Vet Review', 'Awaiting Lab Test', and 'Finalized & Prescribed'. 3. Save consultation.	**Expected Clinical Outcome:** Grid renders in clean 2x2 symmetry with zero black blob radio dots. Selected badge highlights with distinct clinical accent color and persists upon modal save.

TC-07: Grouped Diagnosis Selection via Species Optgroups

Audit Result: [PASS]

Clinical Objective: Verify hierarchical species grouping in the Clinical Override dropdown.	**Preconditions:** Open Patient Case Review modal.
Execution Steps: 1. Click the 'Verified Diagnosis / Clinical Override' dropdown. 2. Inspect grouping structure. 3. Select 'Other / Custom Dermatological Condition...'.	**Expected Clinical Outcome:** Dropdown displays distinct optgroups: Canine Protocols (Aso), Feline Protocols (Pusa), Avian Protocols (Ibon), Rabbit Protocols (Kuneho). Selecting 'Other' reveals a custom text input field.

TC-08: Attending Veterinarian Healing Stage Evaluation Cards

Audit Result: [PASS]

Clinical Objective: Confirm 4-tier clinical healing stage cards render with color accents and micro-indicators.	**Preconditions:** Log in as Attending Veterinarian and open 'Follow-up Recheck' modal.
Execution Steps: 1. Inspect Healing Stage Evaluation section. 2. Select 'Responding / Improving'. 3. Switch to 'Suspected New Complication'.	**Expected Clinical Outcome:** Replaces legacy dark radio dots with modern color-coded cards (Green, Amber, Orange, Red). Selected card shows active ring and colored micro-indicator.

TC-09: 100% Attending Veterinarian Editable Recommended Treatment Protocol

Audit Result: [PASS]

Clinical Objective: Verify that veterinarians can edit, add, delete, and reset clinical guideline steps.	**Preconditions:** Navigate to the Treatment page for a diagnosed patient.
Execution Steps: 1. Under 'RECOMMENDED TREATMENT PROTOCOL', edit an existing guideline step. 2. Click '+ Add Guideline Step' and input custom dosage. 3. Delete a step using 'Remove'. 4. Click 'Reset to Guidelines'.	**Expected Clinical Outcome:** All steps are fully editable text fields. Adding/removing steps updates in real-time. Reset button restores standard evidence-based clinical defaults.

TC-10: Zero-Scroll Email Prescription Modal & 24-Hour Duplicate Dispatch Guard

Audit Result: [PASS]

Clinical Objective: Verify compact split-layout, email dispatch, auto-save to patient record, and duplicate dispatch prevention.	**Preconditions:** On Treatment page, click 'Email Prescription'.
Execution Steps: 1. Verify modal fits within screen without scrolling (form on left, scaled preview on right). 2. Enter owner email if blank (verifies auto-save). 3. Click 'Send via Clinic Cloud Email'. 4. Attempt sending again immediately.	**Expected Clinical Outcome:** First send succeeds and saves email to IndexedDB. Second identical send displays 24-hour duplicate dispatch warning banner, requiring override checkbox confirmation.

TC-11: Patient Records High-Resolution Photo Inspection Modal

Audit Result: [PASS]

Clinical Objective: Verify clicking any patient lesion thumbnail opens full-screen zoomable lightbox viewer.	**Preconditions:** Navigate to Patients list with completed photo cases.
Execution Steps: 1. Click on the lesion photo thumbnail in the patient card or case modal. 2. Inspect image clarity and metadata. 3. Click Close or tap outside backdrop.	**Expected Clinical Outcome:** Opens dedicated ImageViewerModal displaying high-resolution uncompressed image with patient name, diagnosis, and timestamp overlay. Closes cleanly on Esc or backdrop click.

TC-12: End-to-End Clinical Case Lifecycle Progression

Audit Result: [PASS]

Clinical Objective: Validate entire workflow from initial patient intake to final discharge.	**Preconditions:** Complete clinic setup with online or offline database.
Execution Steps: 1. New Patient Intake -> 2. AI Dermatological Scan -> 3. Attending Vet Review & Protocol Customization -> 4. Prescription Print / Email -> 5. Follow-up Recheck & Discharge.	**Expected Clinical Outcome:** Seamless state transition across all views with complete data integrity in IndexedDB and cloud audit trail. No data loss or unhandled exceptions.

5. Clinical Sign-off & Institutional Verification Log

This sign-off sheet certifies that all twelve system verification test cases have been thoroughly conducted, analyzed, and approved according to institutional veterinary software benchmarks.

Role / Stakeholder	Name & Professional Title	Evaluation Verdict	Signature & Date
Lead Developer	Russell / Benedict Dela Cruz Lead Software Engineer	VERIFIED & PASSED All 12 Test Cases Validated	_____ Date: Nov 09, 2026
Attending Clinician	Licensed Doctor of Veterinary Medicine Clinical Field Consultant	CLINICALLY APPROVED Protocols & Safety Enforced	_____ Date: Nov 09, 2026
Thesis Panelist 1	Technical Committee Evaluator Computer Science / IT Faculty	DEFENSE CANDIDATE Architecture Standards Met	_____ Date: Nov 09, 2026
Thesis Panelist 2	Lead Research Advisor Capstone Committee Chair	DEFENSE CANDIDATE Methodology Approved	_____ Date: Nov 09, 2026

Institutional Diagnostic Ethics & Decision Support Boundary:
Vetlink operates exclusively as a clinical diagnostic decision support system under Republic Act No. 8485 (Animal Welfare Act) and standard veterinary licensing laws. In-app artificial intelligence predictions are non-binding and must be reviewed, modified, or corroborated through standard clinical laboratory methods (e.g., deep skin scrapings, acetate tape cytology, dermatophyte test medium culture) by a licensed Doctor of Veterinary Medicine prior to definitive therapeutic administration.